

Understanding Employee Turnover and Enhancing Employee Productivity in the Work

Introduction

The COVID-19 pandemic has brought a significant shift in the way people operate in the organization, with remote work becoming the new standard. While the transition to remote work offers various advantages, it has also presented various challenges for an organization¹. Rising living costs have left many employees feeling underpaid, and the option to work remotely has made it easier for them to explore better-paying jobs outside their local area.

The trend of employees leaving their jobs began in 2021, right after the COVID-19 pandemic, and the trend continued. In 2022, over 50 million American workers made the decision to leave their jobs. This is the highest turnover rate in the history of the Job Openings and Labor Turnover Survey (JOLTS)²—a dataset that has been tracked since its inception in 2001.

The idea of this project is to address these challenges by using the power of data analytics to analyze the employees' sentiment toward the company, identify factors contributing to turnover, and develop data-driven strategies to mitigate them.

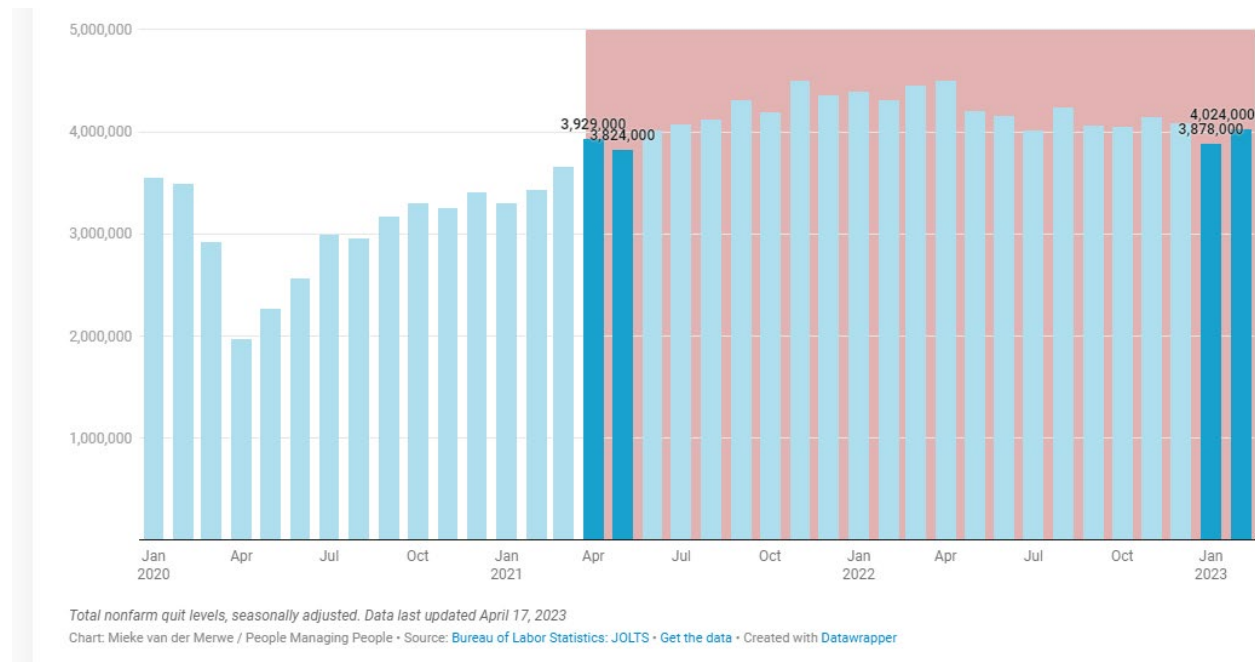


Fig: Number of quits from Jan 2020 to Jan 2023²

Data Exploration

Data Collection and Implementation

To comprehensively understand the sentiments of the employees within an organization and effectively address the turnover and productivity concerns, multifaceted data collection is essential.

For this, I will be focusing on four major data collection domains including turnover data, HR Records, Chat Data from employee's email and chat logs, and Employee Satisfaction Surveys

1. Turnover Data:

Source: Utilizing historical data on employee turnover, including the reason for leaving, exit surveys, and HR records.

Implementation: Organizations can maintain comprehensive records of employees who have left, including their reasons for departure, job roles, compensation and benefits, and their perception of work-life balance. Exit surveys can be conducted to gather more in-depth insights into their experiences and reasons for leaving.

Reason for using this data: Turnover data will help us to differentiate between short-term and long-term employees who are leaving the organization. This is one of the crucial data points that need to be collected because it provides insights into the specific employee categories that are opting to depart. This distinction is valuable as it enables the organization to tailor its strategies for reducing turnover with precision. By understanding which employee segments are departing, we can direct our efforts more effectively, addressing the unique drivers behind turnover in each group.

2. HR Records:

Source: Utilizing HR records, including salary information, promotion history, and personnel files.

Implementation: HR departments typically maintain detailed records of employee information, making it feasible to extract data related to compensation, promotions, and tenure. These records can be helpful in assessing the impact of salary structures and career progression on turnover and productivity.

Reason for this data: The HR Records provide a comprehensive view of employees, including their job history, compensation details, salary information, promotion, and years worked in the organization. By analyzing this data, organizations can pinpoint the turnover patterns and use these patterns to identify at-risk employee segments and use tailored retention strategies. Additionally, the organization can also use the HR records to set the benchmark against industry standards.

3. Chat Data:

Source: Using data from communication mediums within the organization, such as emails and chat logs.

Implementation: By analyzing email and chat communication, organizations can gain insights into employee sentiments and pinpoint areas of concern or dissatisfaction. Natural language processing (NLP) tools can be employed to perform sentiment analysis on textual data, helping to identify recurring themes and emotions.

Reason for this data: Employee chat data offers real-time insights that will help identify the recurring concerns and emotional tones of employees in conversations. By identifying the problem in real-time, the organization can use the proactive approach to take immediate corrective action and prevent the potential turnover from escalating.

4. Employee Surveys:

Source: Getting detailed surveys from employees, covering various aspects of their roles, workloads, compensation, work flexibility, and work-life balance.

Implementation: Conducting regular employee surveys allows organizations to collect direct feedback on various aspects of the workplace. Surveys can be distributed electronically, and results can be analyzed to understand employee sentiments and identify specific issues or trends. Surveys may include questions related to job satisfaction, workload, work-from-home experiences, compensation satisfaction, and overall job engagement.

Reason for this data: Employee Surveys are a direct channel for employees to voice their opinion and concerns about their roles and work environment. The feedback from surveys covers a wide range of topics, including job satisfaction, workload, compensation, and overall job engagement. This data is quantifiable and can be used for statistical analysis to identify patterns and trends.

Data Analysis

Once all the required data is collected, the next phase involves analyzing the data to understand the underlying reason for the employee turnover rate. This section serves as a critical bridge between the raw data collected from various sources and the actionable insights that can bring meaningful insights. Below are some of the analysis techniques that can be implemented to get the insights from the collected data:

Sentiment Analysis: We can employ advanced sentiment analysis techniques to process and analyze survey responses, communication data, and feedback. This approach will provide us with

quantifiable insights into the sentiment among employees. It also enables us to categorize sentiments as positive, negative, or neutral, and identify the key themes and emotional tone within the data.

Turnover Analysis: Using robust software packages like R and Python with libraries such as pandas and NumPy will be essential for conducting turnover analysis. With the help of these packages, we will be able to analyze historical turnover data. Additionally, visualization tools like Matplotlib and Tableau can also aid in presenting findings effectively.

By examining trends and patterns, we can uncover common factors contributing to employee departures. This analysis will provide information on the 'why' behind turnover, allowing us to develop strategies that directly address these factors.

Correlation Studies: Analytical tools, including R, Python, and dedicated statistical software can be employed for correlation studies. Investigating potential correlations between turnover factors, compensation, and satisfaction metrics will be a crucial part of the analysis. By quantifying these relationships, we can gain insights into the interdependencies that may not be immediately apparent.

Predictive Modeling: Machine Learning libraries within Python and R can be used to analyze the sentiment of the employee. Using predictive modeling tools, we will be able to develop sophisticated models that can anticipate potential turnover risks and identify high-potential employees. These tools can also help us make educated guesses about who might leave the company and the organization can use that information to mitigate employee turnover.

Once we have gathered the data from sentiment analysis and turnover analysis, the next step involves predictive modeling. Predictive modeling is important in determining the significance of various variables within the dataset. By using predictive modeling, we can identify which variable has more importance and which has lesser importance. Using only the relevant variable will help us to focus our attention on the key drivers affecting turnover and retention, ultimately guiding the development of more persistent and effective strategies for mitigating turnover. Additionally, a decision tree will be created to see the factors that contribute to employee turnover.

Knowledge Transfer

In this section, we look into how these insights can be harnessed to contribute to a deeper understanding of employee sentiments and advance our goal of addressing turnover and enhancing productivity within the organization.

1. Understanding Employee Sentiments

By categorizing sentiments as positive, negative, or neutral and identifying key themes and emotional tones, we gain a comprehensive understanding of how employees perceive their workplace. This knowledge contributes to understanding the factors influencing employee satisfaction, motivation, and engagement, ultimately aiding in crafting strategies to improve workplace sentiment.

2. Identifying Turnover Factors

Statistical analyses uncover commonalities and relationships among variables, shedding light on why employees leave and how various factors interconnect. This knowledge informs the development of targeted retention strategies by identifying the root causes of turnover. It empowers organizations to proactively address these factors to reduce turnover rates.

3. Enhancing Productivity

Predictive models utilize historical data to forecast future turnover risks and identify employees with significant growth potential. This knowledge aids in resource allocation and strategic decision-making, allowing organizations to focus efforts on retaining valuable talent and fostering productivity.

4. Informed Decision-Making

By combining insights from sentiment analysis, turnover analysis, and predictive modeling, organizations gain a holistic view of their workforce and the factors that impact turnover and productivity. This knowledge forms the foundation for crafting evidence-based strategies, addressing workplace issues, and creating a more conducive and productive work environment.

Finally, the insights gathered from the predictive model and decision tree will be shared with the HR team and the senior management with the aim of implementing measures to proactively reduce employee turnover. This valuable information has the potential to result in substantial, long-term cost savings for the organization. By acting upon these insights, the company can avoid significant expenses associated with recruiting and training new employees when turnover occurs.

References:

1. Lorie Konish (Sep 14, 2022). 55% of Workers Say Their Pay Isn't Keeping Up with The Inflation. CNBC. <https://www.cnbc.com/2022/09/14/half-of-working-americans-say-their-pay-isnt-keeping-up-with-inflation.html>
2. People Managing People (2023). Employee Retention Statistics You Should Know in 2023. <https://peoplemanagingpeople.com/employee-lifecycle/employee-retention-statistics/#:~:text=In%202022%2C%20a%20record%2050.6,which%20dates%20back%20to%202001>.